LAB EXPERIMENT RECORD

Set 16: Customer Demographics Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in $)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

1. Bar Chart

Aim

To create a bar chart to visualize the distribution of customer ages and label the axes and title the chart.

R Code

```
customer_age <- c(28, 35, 42)

customer_id <- c("Customer 1", "Customer 2", "Customer 3")

barplot(customer_age,

    names.arg = customer_id,

    main = "Distribution of Customer Ages",
```

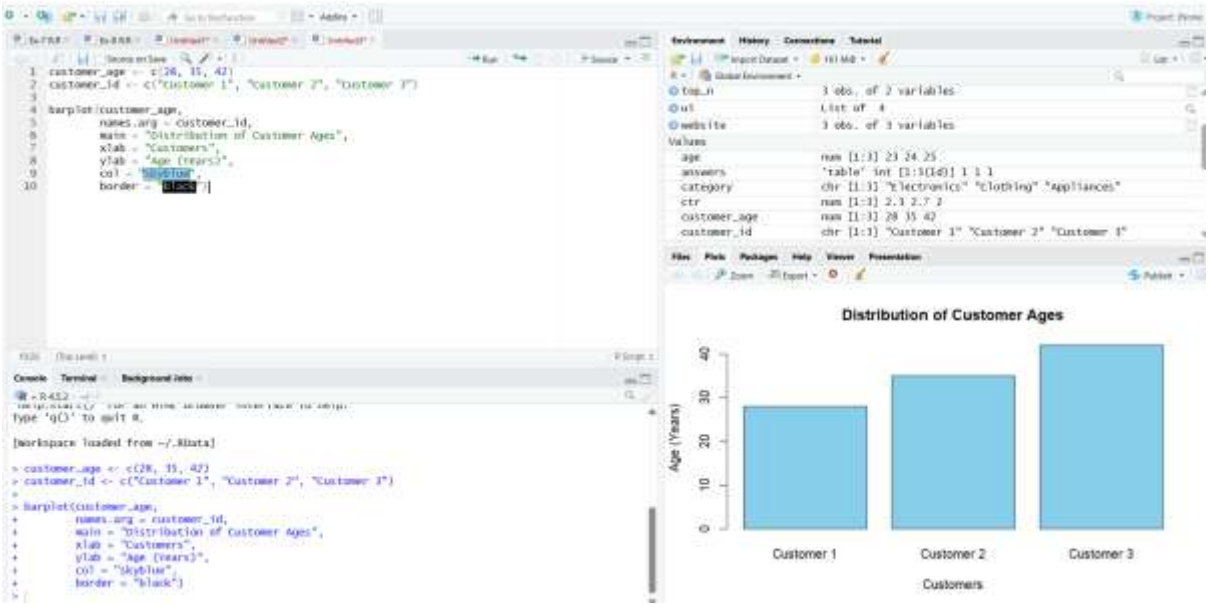
xlab = "Customers",

ylab = "Age (Years)",

col = "skyblue",

border = "black")

Output



Observation

Customer 1 is 28 years old, Customer 2 is 35 years old, and Customer 3 is 42 years old. Customer 3 is the oldest, while Customer 1 is the youngest.

2. Pie Chart

Aim

To generate a pie chart representing the distribution of customers by gender.

R Code

gender_count <- c(2, 1)

gender_labels <- c("Female", "Male")

pie(gender_count,

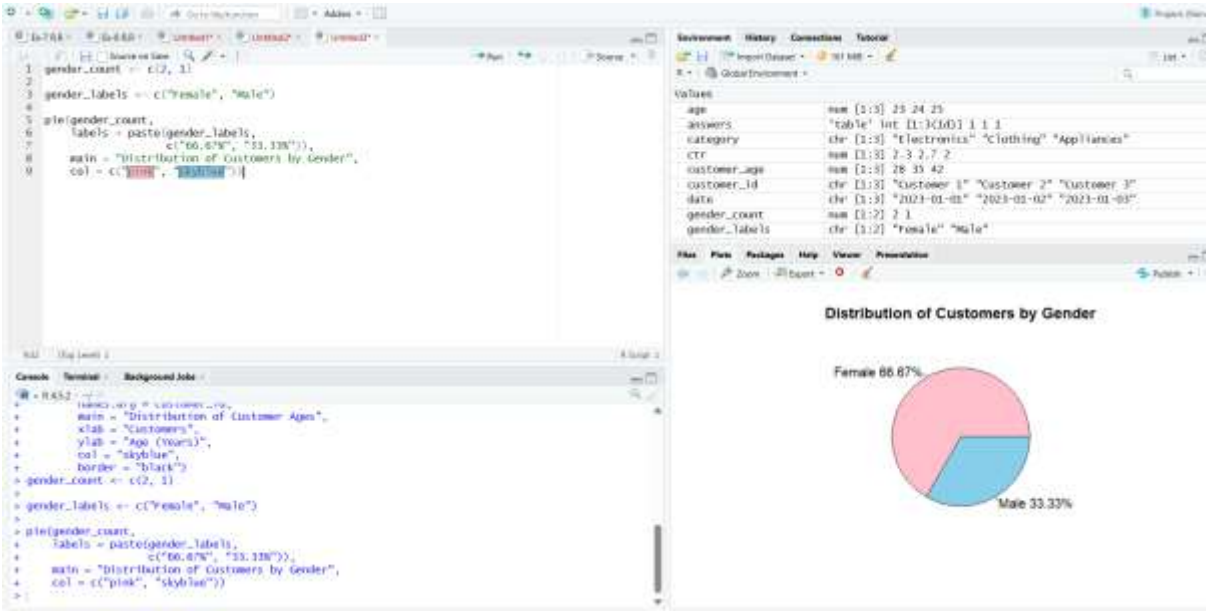
labels = paste(gender_labels,

c("66.67%", "33.33%")),

main = "Distribution of Customers by Gender",

col = c("pink", "skyblue"))

Output



A pie chart showing the gender distribution of customers, with 66.67% female customers and 33.33% male customers.

Observation

There are two female customers and one male customer. Female customers represent 66.67% of the dataset, while male customers represent 33.33%.

3. Customer Demographics Table

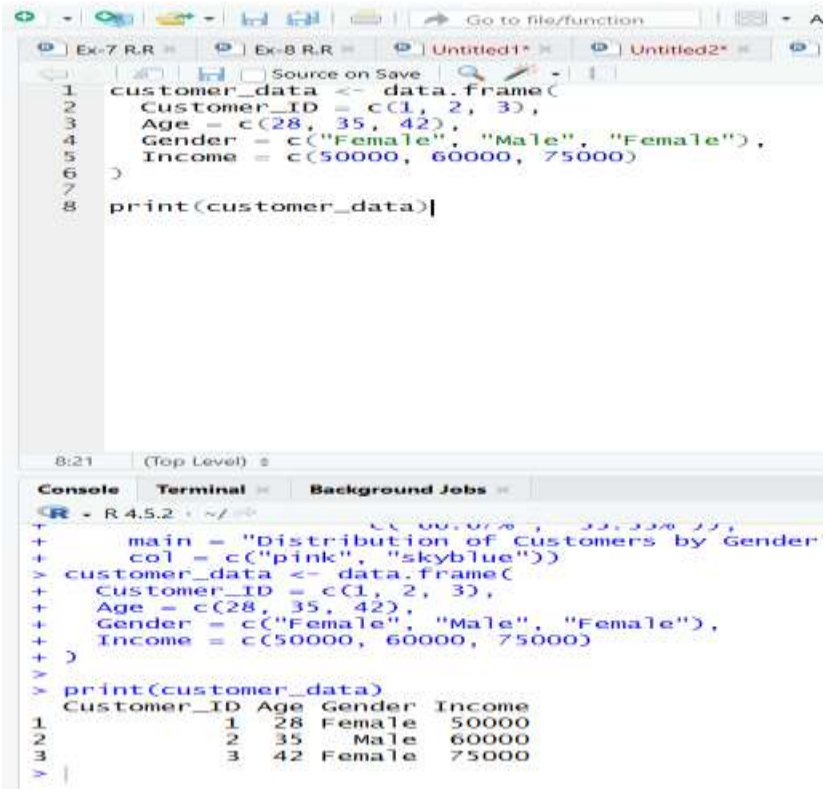
Aim

To create a table displaying the demographic information of each customer, including customer ID, age, gender, and income.

R Code

```
customer_data <- data.frame(
    Customer_ID = c(1, 2, 3),
    Age = c(28, 35, 42),
    Gender = c("Female", "Male", "Female"),
    Income = c(50000, 60000, 75000)
)
print(customer_data)
```

Output



Observation

The table clearly displays the demographic details of all three customers. Customer 3 has the highest income of $75,000, while Customer 1 has the lowest income of $50,000.

4. Tableau Dashboard

Aim

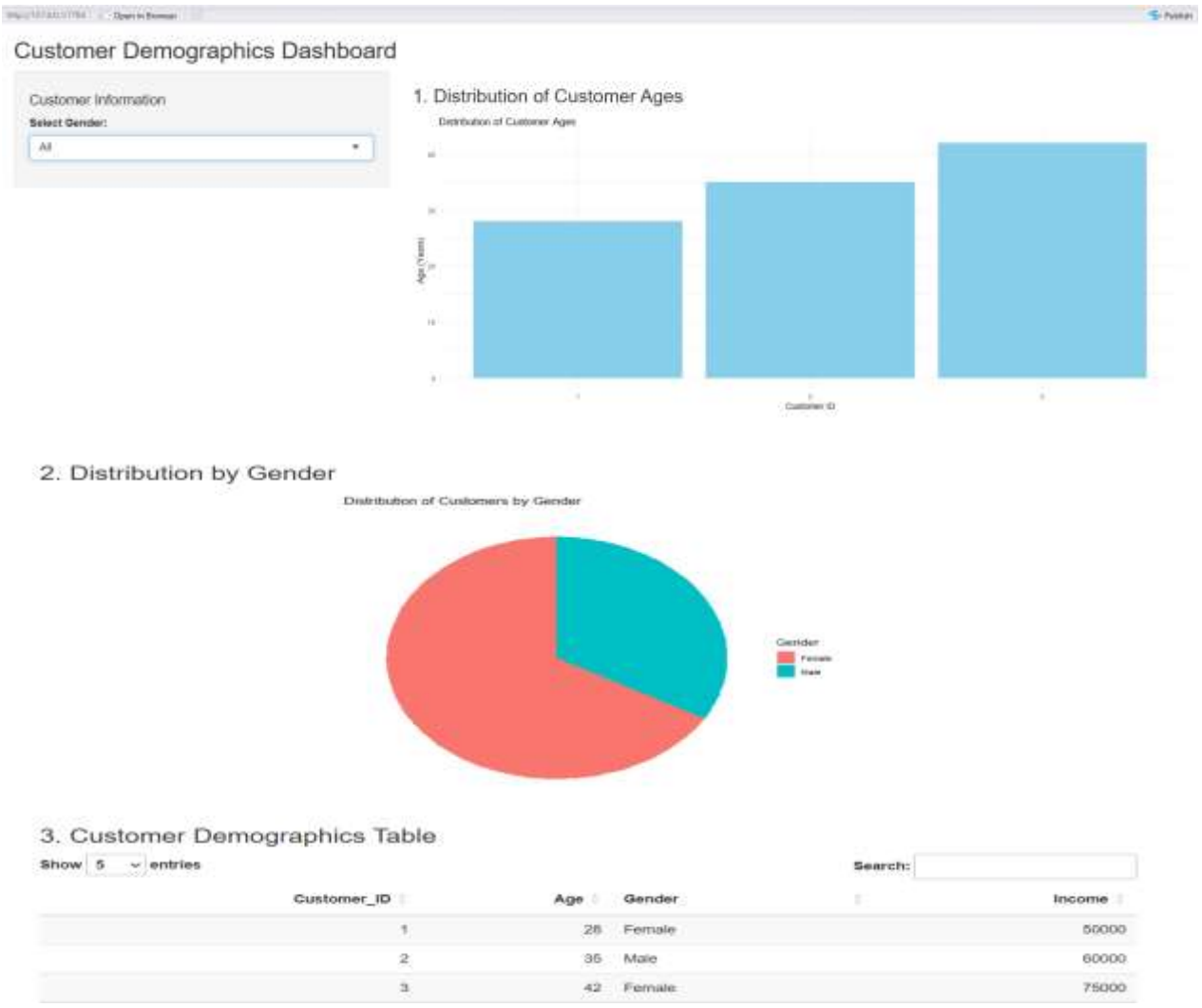
To develop an interactive Tableau dashboard combining the bar chart, pie chart, and customer demographics table for exploring customer demographic information.

Tableau Procedure

1. Open Tableau.
2. Import the Customer Demographics dataset.
3. Create a bar chart using Customer ID and Age.
4. Add suitable chart titles and axis labels.
5. Create a pie chart using Gender and the count of Customer ID.
6. Display gender proportions using percentage labels.
7. Create a table containing Customer ID, Age, Gender, and Income.

8. Select Dashboard → New Dashboard.
9. Drag the bar chart into the dashboard.
10. Drag the pie chart into the dashboard.
11. Drag the customer demographics table into the dashboard.
12. Arrange all charts and the table neatly.
13. Add suitable titles and labels.
14. Add filters if required for interactive exploration.
15. Save the dashboard.

Output



An interactive Tableau dashboard displaying the distribution of customer ages, gender proportions, and demographic information for all customers.

Result

The customer demographics dataset was successfully analyzed using a bar chart, pie chart, and table in R. A Tableau dashboard was designed to provide an interactive view of customer ages, gender distribution, and income details.